1/26/2022

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M168_Notes_2.2

2.2 Systems of Linear Equations: Unique Solutions

You may:

- Interchange any two of the equations.
- Multiply an equation by a non-zero constant.
- Add a multiple of one equation to another.

$$\boxed{\begin{aligned} 2x - y &= 6 \\ x + 2y &= -2 \end{aligned}}$$

$$\begin{array}{cc} x & y \\ \downarrow & \downarrow \end{array}$$

$$\left[\begin{array}{cc|c} 2 & -1 & 6 \\ 1 & 2 & -2 \end{array}\right]$$

messy

"Work"

$$\begin{array}{cc} x & y \\ \downarrow & \downarrow \end{array}$$

$$\left[\begin{array}{cc|c} 1 & 0 & a \\ 0 & 1 & b \end{array}\right], \quad x = a, \quad y = b$$

clean

AUGMENTED MATRIX form

Rules are

- Interchange any of the rows equations.
- Multiply a row by a non-zero constant.
- Add a multiple of one row to another.

The goal is to get the augmented matrix into row-reduced echelon form (RREF).

What is RREF form?

1. All rows consisting entirely of zeros are at the bottom of the matrix
2. The first non-zero entry in any row is a 1 (called a leading 1)
3. In any two successive non-zero rows the leading 1 in the lower row lies to the right of the leading 1 in the upper row.
4. If a column contains a leading 1, then the rest of the entries in that column are 0's.

$$\begin{matrix} 0 & 1 & 0 & 0 \\ a & 0 & 1 & 0 \\ a & b & 0 & d \end{matrix}$$

$(1, 2, -2)$

$-(1, -\tfrac{1}{2}, 3)$

$= (1-1, 2-(-\tfrac{1}{2}), -2-3)$

PIVOT on an element – make the element a 1 and the rest of the entries in that column 0's.

Gauss-Jordan Method – Pivot until the augmented matrix is in RREF form.

$$\begin{aligned} 2x - y &= 6 \\ x + 2y &= -2 \end{aligned} \Rightarrow \left[\begin{array}{cc|c} 2 & -1 & 6 \\ 1 & 2 & -2 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & a \\ 0 & 1 & b \end{array}\right]$$

$$\left[\begin{array}{cc|c} 2 & -1 & 6 \\ 1 & 2 & -2 \end{array}\right] \xrightarrow[(2)]{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{cc|c} 1 & -\tfrac{1}{2} & 3 \\ 1 & 2 & -2 \end{array}\right] \xrightarrow[(4)]{R_2 \to R_2 - R_1} \left[\begin{array}{cc|c} 1 & -\tfrac{1}{2} & 3 \\ 0 & 2-(-\tfrac{1}{2}) & -2-3 \end{array}\right] \xrightarrow{\text{simplify}} \left[\begin{array}{cc|c} 1 & -\tfrac{1}{2} & 3 \\ 0 & 2.5 & -5 \end{array}\right]$$

$$\xrightarrow[(2)]{R_2 \to \frac{R_2}{2.5}} \left[\begin{array}{cc|c} 1 & -\tfrac{1}{2} & 3 \\ 0 & 1 & -2 \end{array}\right] \xrightarrow[(4)]{R_1 \to R_1 + \frac{R_2}{2}} \left[\begin{array}{cc|c} 1 & 0 & 3-1 \\ 0 & 1 & -2 \end{array}\right] \xrightarrow{\text{simp.}} \begin{array}{c} \begin{array}{cc} x & y \end{array} \\ \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -2 \end{array}\right] \end{array}$$

$x = 2$

$y = -2$

$$\begin{aligned} x + y + z &= 192 \\ y + z &= x \\ x &= 3z \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 192 \\ -1 & 1 & 1 & 0 \\ 1 & 0 & -3 & 0 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 96 \\ 0 & 1 & 0 & 64 \\ 0 & 0 & 1 & 32 \end{array}\right]$$

$$\boxed{\begin{aligned} x &= 96 \\ y &= 64 \\ z &= 32 \end{aligned}}$$